



NORTH SYDNEY GIRLS HIGH SCHOOL

HSC Mathematics Extension 1

Assessment Task 3

Term 2, 2013

Name: _____ Mathematics Class: _____

Student Number _____

Time Allowed: 50 minutes + 2 minutes reading time

Total Marks: 40

Instructions:

- Attempt all eight questions.
- Show all necessary working in questions 6 to 8.
- Each question will be collected separately. Submit a blank booklet if you do not attempt a question.
- A table of standard integrals is provided at the back of this paper.

Question	I q 1	I q 2, 3, 5	I Q 4	6	7a	7bc	8a	8b	Total
PE3			/1			9			/10
HE4				/12	/2				/14
HE6		/3					/3		/6
HE7	/1							/9	/10
									/40

Section I

5 marks

Attempt Questions 1 – 5

Allow about 7 minutes.

Use multiple choice answer sheet for questions 1 -5

1. What is the value of $\lim_{x \rightarrow 0} \frac{\sin 7x}{5x}$?

(A) $\frac{7}{5}$

(B) 1

(C) $\frac{5}{7}$

(D) 0

2. What is the derivative of $y = \cos^{-1}(2x)$?

(A) $\frac{-1}{2\sqrt{1-4x^2}}$

(B) $\frac{1}{2\sqrt{1-4x^2}}$

(C) $\frac{-2}{\sqrt{1-4x^2}}$

(D) $\frac{2}{\sqrt{1-4x^2}}$

3. Which of the following is the inverse function of $f(x) = 3 - \frac{1}{x+2}$?

(A) $f^{-1}(x) = -2 - \frac{1}{3-x}$

(B) $f^{-1}(x) = 2 - \frac{1}{3-x}$

(C) $f^{-1}(x) = -2 + \frac{1}{3-x}$

(D) $f^{-1}(x) = 2 + \frac{1}{3-x}$

4. When the substitution $u = \sin x$ is used, $\int_0^{\frac{\pi}{2}} \cos x \sin^4 x \, dx$ is equivalent to

(A) $\int_0^1 u^4 du$

(B) $-\int_0^1 u^4 du$

(C) $\int_0^1 u(1-u^2)^2 du$

(D) $-\int_0^1 u(1-u^2)^2 du$

5. By using symmetry, what is the value of $\int_{-a}^a \cos^{-1} x \, dx$ where $-1 \leq a \leq 1$?

(A) 0

(B) $\frac{a\pi}{2}$

(C) $a\pi$

(D) $2a\pi$

End of Section I

Section II

30 marks

Attempt Questions 6 - 8

Allow about 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing paper is available.

In Questions 6-8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (12 Marks)

Use a Separate writing booklet

- (a) Evaluate $\int_0^4 \frac{1}{16+x^2}$. **2**
- (b) Differentiate $\sin^{-1}\left(\frac{a}{x}\right)$ with respect to x . **2**
- (c) (i) What is the domain of $y = \sin^{-1}(2x-1)$? **1**
- (ii) Sketch the graph of $y = \sin^{-1}(2x-1)$ **2**
- (d) (i) If $\theta = \tan^{-1} A + \tan^{-1} B$ then show that $\tan \theta = \frac{A+B}{1-AB}$ **2**
- (ii) Hence solve the equation $\tan^{-1} 3x + \tan^{-1} 2x = \frac{\pi}{4}$ **3**

End of Question 6

Question 7 (11 Marks)

Use a Separate writing booklet.

(a) Find $\lim_{x \rightarrow 1} \frac{(x^2 - 1) + \tan(x - 1)}{x - 1}$.

2

- (b) A furnace is turned off and it begins cooling in a room at a constant temperature 22°C .

At time t minutes its temperature T decreases according to the equation

$$\frac{dT}{dt} = -k(T - 22) \text{ where } k \text{ is a positive constant.}$$

The initial temperature of the furnace is 120°C and it cools to 90°C after 20 minutes.

- (i) Verify that $T = 22 + Ae^{-kt}$ is a solution to the differential equation, where A is a constant.

1

- (ii) Find the values of both A and k .

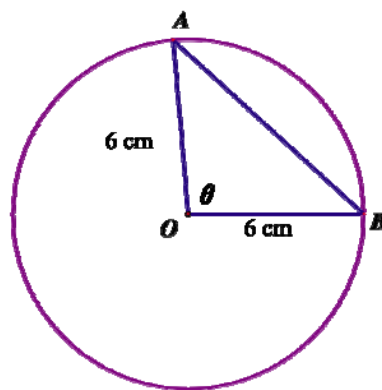
2

- (iii) How long will it take for the heater to cool to 45°C .
Give your answer to the nearest minute.

2

- (c) A and B are points on the circumference of a circle with centre O and radius 6 cm.
Let $\angle AOB = \theta$.

θ is increasing at a constant rate of 0.2 radians per second.



- (i) Let S be the area of the minor segment created by the chord AB in square centimetres. Show that

2

$$\frac{dS}{d\theta} = 18(1 - \cos \theta)$$

- (ii) Find the rate of change of S when $\theta = \frac{2\pi}{3}$

2

End of Question 7

Question 8 (12 Marks)

Use a Separate writing booklet

- (a) Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 2x \, dx$ **3**
- (b) Consider the function $f(x) = \frac{1}{3}[(x-1)^2 + 5]$.
- (i) Sketch the parabola $y = f(x)$ showing clearly the coordinates of the vertex and any intercepts with the axes. **1**
- (ii) What is the largest domain containing $x = 2$ for which the function has an inverse function $f^{-1}(x)$? **1**
- (iii) State the domain of $f^{-1}(x)$ **1**
- (iv) Find the coordinates of the points of intersection of the two curves $y = f^{-1}(x)$ and $y = f(x)$ **2**
- (v) Sketch the graph of $y = f^{-1}(x)$ **2**
- (vi) Let p be a real number such that $p < 1$ **2**
Find $f^{-1}(f(p))$.

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Trial HSC Examination – Mathematics Extension 1 2012

Multiple Choice Answer Sheet

Name _____

Completely fill the response oval representing the most correct answer.

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

Section I – Multiple Choice

Q1 A

$$\lim_{x \rightarrow 0} \frac{\sin 7x}{5x} = \frac{7}{5} \lim_{x \rightarrow 0} \frac{\sin 7x}{7x} = \frac{7}{5} \times 1$$

Q2 C

$$y = \cos^{-1}(2x)$$

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1-(2x)^2}} \times 2 = \frac{-2}{\sqrt{1-4x^2}}$$

Q3 C

$$y = 3 - \frac{1}{x+2}$$

The inverse is

$$x = 3 - \frac{1}{y+2}$$

$$\frac{1}{y+2} = 3 - x$$

$$y+2 = \frac{1}{3-x}$$

$$y = -2 + \frac{1}{3-x}$$

Q4 A

$$\int_0^{\frac{\pi}{2}} \cos x \sin^4 x dx \quad x = \frac{\pi}{2}$$

$$u = \sin x$$

$$\frac{du}{dx} = \cos x \quad u = \sin \frac{\pi}{2} = 1$$

$$x = 0$$

$$u = \sin 0 = 0$$

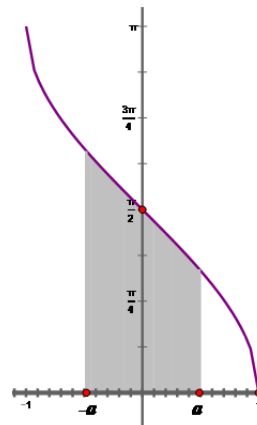
$$\int_0^1 \cos x u^4 \frac{du}{\cos x}$$

$$\int_0^1 u^4 du$$

Q5 C

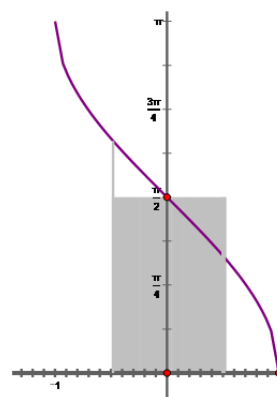
$$\int_{-a}^a \cos^{-1} x dx$$

is this area



because of the symmetry of $\cos^{-1} x$

the area is also equivalent to



The area is a rectangle

$$A = 2a \times \frac{\pi}{2} = a\pi$$

$$\int_{-a}^a \cos^{-1} x dx = a\pi$$

Section II

Question 6

a)

$$\int_0^4 \frac{1}{16+x^2} dx = \left[\frac{1}{4} \tan^{-1} \frac{x}{4} \right]_0^4$$

$$= \left(\frac{1}{4} \tan^{-1} 1 \right) - (0)$$

$$= \frac{1}{4} \times \frac{\pi}{4} = \frac{\pi}{16}$$

b)

$$\frac{d}{dx} \sin^{-1} \frac{a}{x} = \frac{1}{\sqrt{1-\frac{a^2}{x^2}}} \times \left(\frac{-a}{x^2} \right)$$

$$= \frac{-a}{x^2 \sqrt{1-\frac{a^2}{x^2}}}$$

$$= \frac{-a}{x \sqrt{x^2 - a^2}}$$

OR

$$= \frac{-a}{\sqrt{x^4 - a^2 x^2}}$$

c)

Domain

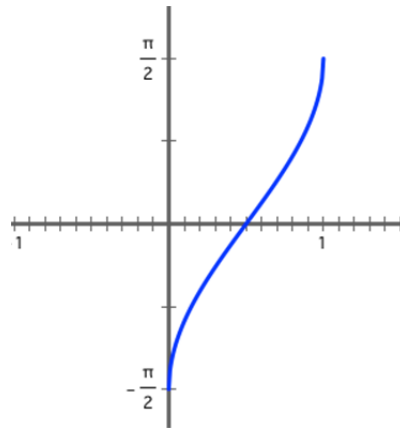
$$-1 \leq 2x-1 \leq 1$$

$$0 \leq 2x \leq 2$$

$$0 \leq x \leq 1$$

Range

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



d)i)

$$\text{let } p = \tan^{-1} A$$

$$\therefore A = \tan p$$

$$\text{and } q = \tan^{-1} B$$

$$\therefore B = \tan q$$

$$\therefore \theta = p + q$$

$$LHS = \tan \theta$$

$$= \tan(p + q)$$

$$= \frac{\tan p + \tan q}{1 - \tan p \tan q}$$

$$= \frac{A + B}{1 - AB}$$

$$= RHS$$

d)ii)

$$\tan^{-1} 3x + \tan^{-1} 2x = \frac{\pi}{4}$$

$$\therefore \tan \frac{\pi}{4} = \frac{3x + 2x}{1 - 6x^2}$$

$$1 = \frac{5x}{1 - 6x^2}$$

$$1 - 6x^2 = 5x$$

$$0 = 6x^2 + 5x - 1$$

$$0 = (6x - 1)(x + 1)$$

$$x = \frac{1}{6} \quad x = -1$$

$$\text{But when } x = -1$$

$$\tan^{-1}(-3) + \tan^{-1}(-2) \neq \frac{\pi}{4}$$

$$\therefore x = \frac{1}{6}$$

Question 7

a)

$$\lim_{x \rightarrow 1} \frac{(x^2 - 1) + \tan(x - 1)}{x - 1}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} + \lim_{x \rightarrow 1} \frac{\tan(x-1)}{x-1}$$

$$\lim_{x \rightarrow 1} (x+1) + \lim_{x \rightarrow 1} \frac{\tan(x-1)}{x-1}$$

$$= (1+1) + 1 = 3$$

b)i)

$$\text{If } T = 22 + Ae^{-kt}$$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$\frac{dT}{dt} = -k(T - 22)$$

$$\text{note : } T - 22 = Ae^{-kt}$$

b)ii)

$$\text{When } t = 0 \quad T = 120^\circ$$

$$120 = 22 + Ae^{-k(0)}$$

$$120 = 22 + A$$

$$98 = A$$

$$90 = 22 + 98e^{-20k}$$

$$\frac{68}{98} = e^{-20k}$$

$$\ln\left(\frac{68}{98}\right) = -20k$$

$$k = \frac{\ln\left(\frac{68}{98}\right)}{-20} = k = 0.0182729...$$

b)iii)

$$\text{When } T = 45^\circ$$

$$45 = 22 + 98e^{-kt}$$

$$\frac{23}{98} = e^{-kt}$$

$$\ln\left(\frac{23}{98}\right) = -kt$$

$$\frac{\ln\left(\frac{23}{98}\right)}{-k} = t$$

$$\frac{-20 \ln\left(\frac{23}{98}\right)}{\ln\left(\frac{68}{98}\right)} = t$$

$$t = 79.32327265 \text{ min}$$

$$t = 79 \text{ min}$$

$$\frac{d\theta}{dt} = 0.2 \text{ rad/s} \quad \text{and} \quad r = 6 \text{ cm}$$

$$S = \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$$

$$S = \frac{1}{2}(6)^2\theta - \frac{1}{2}(6)^2\sin\theta$$

$$S = 18\theta - 18\sin\theta$$

$$\frac{dS}{d\theta} = 18\theta - 18\cos\theta$$

$$\frac{dS}{d\theta} = 18(1 - \cos\theta)$$

$$\text{Find } \frac{dS}{dt} \quad \text{when} \quad \theta = \frac{2\pi}{3}$$

$$\frac{dS}{dt} = \frac{dS}{d\theta} \times \frac{d\theta}{dt}$$

$$\frac{dS}{dt} = 0.2 \times 18 \left(1 - \cos\frac{2\pi}{3}\right)$$

$$\frac{dS}{dt} = 5.4 \text{ cm}^2/\text{s}$$

Question 8

a)i)

$$\int_0^{\frac{\pi}{2}} \cos^2 2x \, dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 + \cos 4x \, dx$$

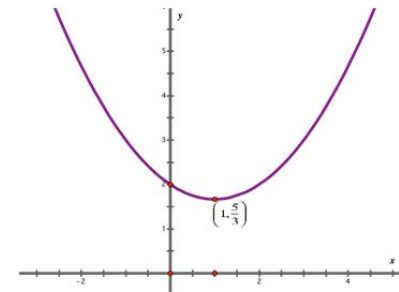
$$= \frac{1}{2} \left[x + \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{2} - 0 \right) - (0 - 0) \right]$$

$$= \frac{\pi}{4}$$

b)i)

$$\text{Vertex is } \left(1, \frac{5}{3}\right)$$



b)ii)

Largest domain is $x \geq 1$

b)iii)

Domain is $x \geq \frac{5}{3}$

b)iv)

The inverse function $y = f^{-1}(x)$ will intersect $y = f(x)$ on the line $y = x$

$$\therefore x = \frac{1}{3} \left[(1-x)^2 + 5 \right]$$

$$3x = 1 - 2x + x^2 + 5$$

$$0 = x^2 - 5x + 6$$

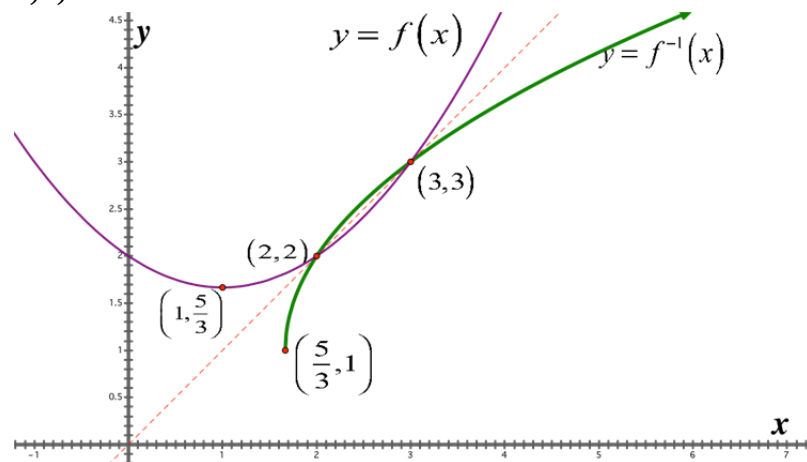
$$0 = (x-3)(x-2)$$

$$x = 2 \quad x = 3$$

$\therefore$ Points of intersection are

(2, 2) and (3, 3)

b)v)



b)vi)

let q be the x -value such that

$$f(p) = f(q) \text{ and } q \geq 1$$

Then p and q are equidistant from $x = 1$

p is a distance of $1 - p$ to the left of $x = 1$

$\therefore q$ is a distance of $1 - p$ to the right of $x = 1$

$$\therefore q = 1 + 1 - p$$

$$q = 2 - p$$

$$\therefore f(p) = f(2 - p)$$

$$\therefore f^{-1}(f(p)) = f^{-1}(f(2 - p)) = 2 - p$$

